Small-time controllability without polynomial reachable neighborhoods

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Abstract

We construct a polynomial control-affine system in five dimensions, with two controls in $[-1,1]$, that is small-time locally controllable at the origin but admits no polynomial lower bound for its reachable neighborhoods. Every point on the negative fourth axis reachable in time at most t has distance less than $t^{11}e^{-1/(6t)}/4$ from the origin. Nevertheless, exact positive and negative fourth-coordinate loops, followed by linear steering in the other coordinates, reach a full neighborhood for each $t > 0$. The construction combines a critical quartic identity with a logarithmic estimate and concentration near its equality profile. It gives a negative answer to the polynomial-growth question for analytic systems with bounded measurable controls.

1 The growth question and a counterexample

Consider a control-affine system near $0 \in \mathbb{R}^n$,

$$\dot{x} = X_0(x) + \sum_{j=1}^{m} u_j X_j(x), \qquad x(0) = 0, \qquad u \in [-1,1]^m.$$

Controls are measurable, and solutions are absolutely continuous and satisfy the equations almost everywhere. Write $\mathcal{R}_{\le t}(0)$ for the endpoints reached in a duration at most t. Small-time local controllability at the origin means that 0 is an interior point of $\mathcal{R}_{\le t}(0)$ for every sufficiently small $t > 0$. A stronger quantitative property is the existence of $C, T > 0$ and an integer $N \ge 1$ such that

$$B(0, Ct^N) \subset \mathcal{R}_{\le t}(0), \qquad 0 < t < T, \tag{1}$$

where balls are Euclidean.

Jafarpour's Conjecture 1.2 in the v7 preprint PDF of [1] asks whether analyticity and small-time local controllability imply this polynomial growth property. The question also appears as AIM Problem 417 for analytic systems with $n \ge 3$ and $X_0(0) = 0$ [2]. The following example satisfies these assumptions and violates (1).

Theorem 1. *On $\mathbb{R}^5$, consider*

$$\begin{aligned} &\dot{x}_1 = u, && \dot{x}_2 = x_1, \quad \dot{x}_3 = x_2, \\ &\dot{x}_4 = x_1^2 x_3^2 - \left(\frac{1}{9} + x_5^2\right) x_2^4, && \dot{x}_5 = v, \end{aligned} \tag{2}$$

with measurable controls $|u|, |v| \le 1$. Its vector fields are polynomial and its drift vanishes at the origin. For every $t > 0$ there is $\delta(t) > 0$ such that

$$B(0, \delta(t)) \subset \mathcal{R}_{\le t}(0). \tag{3}$$

On the other hand, every endpoint $-re_4 \in \mathcal{R}_{\le t}(0)$ *with* $r > 0$ *satisfies*

$$r < \frac{t^{11}}{4} \exp\left(-\frac{1}{6t}\right). \tag{4}$$

Consequently, for every $C, T > 0$ *and every integer* $N \ge 1$*, there is a* $t \in (0, T)$ *for which* (1) *fails.*

The fourth equation is based on the quartic competition studied by Beauchard and Marbach [3, Section 6], with a constant coefficient replaced by $1/9 + x_5^2$. Their Lemma 6.1 proves that $1/9$ is a lower bound for the optimal constant in the associated quartic inequality; Remark 6.2 leaves its exact value open. We prove the critical identity and construct concentrating curves directly below. At the coefficient $1/9$, the fourth displacement of a closed triple-integrator trajectory is nonnegative. An arbitrarily small positive value of x_5^2 permits a negative displacement, but only after concentration on an extremely small scale. This produces controllability without a polynomial reachable ball.

Section 2 proves the bound for all admissible controls. Section 3 constructs a full reachable neighborhood and completes the proof of Theorem 1.

2 A logarithmic estimate for closed trajectories

For any finite time interval, the first, fifth, second and third coordinates of (2) are obtained by successive integration. The fourth is then the integral of a bounded polynomial expression. This gives a unique global absolutely continuous solution for every admissible control and every initial condition. Changing the initial fourth coordinate simply translates that coordinate, since none of the equations depends on x_4. From the zero initial state, for $0 \le s \le t$,

$$|x_1(s)| \le s, \qquad |x_2(s)| \le \frac{s^2}{2}, \qquad |x_3(s)| \le \frac{s^3}{6}, \qquad |x_5(s)| \le s. \tag{5}$$

In particular, $f = x_3$ is C^2, f'' is Lipschitz, and $f''' = u$ almost everywhere.

The estimate needed for the fourth displacement concerns curves with zero endpoint jets. All integrals in this note are Lebesgue integrals. We use integration by parts for absolutely continuous functions [4, Appendix 3.A, Theorems 3.60 and 3.62].

Lemma 2. *Let* $t > 0$ *and let* $f \in C^2([0, t])$ *have Lipschitz second derivative,* $\|f'''\|_{L^\infty} \le 1$*, and*

$$f(0) = f'(0) = f''(0) = f(t) = f'(t) = f''(t) = 0.$$

Set

$$A = \int_0^t f'^4 \, \mathrm{d}s, \qquad q = ff'' + \frac{1}{3} f'^2, \qquad E = \int_0^t q^2 \, \mathrm{d}s.$$

Then

$$E = \int_0^t (ff'')^2 \, \mathrm{d}s - \frac{1}{9} A. \tag{6}$$

If $A > 0$*, then*

$$E \ge \frac{A}{36 \log^2(t^9/(4A))}. \tag{7}$$

Proof. Integration by parts gives

$$\int_0^t ff''f'^2 \, \mathrm{d}s = \frac{1}{3} \int_0^t f(f'^3)' \, \mathrm{d}s = -\frac{1}{3} A.$$

Expanding q^2 proves (6). The initial jets and the third-derivative bound give

$$|f''| \le B := t, \qquad |f'| \le D := \frac{t^2}{2}, \qquad |f| \le S := \frac{t^3}{6}.$$

Assume $A > 0$ and put

$$\varepsilon = \frac{A}{6tBD^2} = \frac{2A}{3t^6}.$$

Since $A \le tD^4 = t^9/16$, we have $0 < \varepsilon \le t^3/24 < S$. Let $c(z) = \max(-\varepsilon, \min(z, \varepsilon))$. Integration by parts yields

$$A_{\mathrm{lo}} := \int_{\{|f|<\varepsilon\}} f'^4 \, \mathrm{d}s = -3 \int_0^t c(f) f'^2 f'' \, \mathrm{d}s \le 3\varepsilon t B D^2 = \frac{A}{2}. \tag{8}$$

Here and below, the truncated compositions are Lipschitz and hence absolutely continuous. For a C^1 function f, its derivative is zero almost everywhere on each level set $\{f = a\}$: points on that set with $f' \neq 0$ are isolated in the level set and form a countable set. At a threshold point with $f' = 0$, the chain rule for a Lipschitz composition follows directly from its Lipschitz bound. Thus the usual chain rules hold almost everywhere for these finitely truncated functions, and no threshold level contributes to an integral containing f'^4.

Define

$$w(z) = \begin{cases} \log(S/\varepsilon), & |z| \le \varepsilon, \\ \log(S/|z|), & \varepsilon < |z| < S, \\ 0, & |z| \ge S. \end{cases}$$

This is a bounded Lipschitz function. On the range of f,

$$0 \le w(f) \le L := \log(S/\varepsilon) = \log \frac{t^9}{4A}.$$

A second integration by parts gives

$$\begin{aligned} \int_0^t q f'^2 w(f) \, \mathrm{d}s &= \frac{1}{3} \int_0^t f w(f) (f'^3)' \, \mathrm{d}s + \frac{1}{3} \int_0^t f'^4 w(f) \, \mathrm{d}s \\ &= -\frac{1}{3} \int_0^t f w'(f) f'^4 \, \mathrm{d}s = \frac{1}{3} A_{\mathrm{hi}}, \end{aligned} \tag{9}$$

where $A_{\mathrm{hi}} = \int_{\{|f|>\varepsilon\}} f'^4 \, \mathrm{d}s$. The boundary terms vanish, and $z w'(z) = -1$ when $\varepsilon < |z| < S$. The level sets $|f| = \varepsilon$ and $|f| = S$ give zero contributions as just explained. By (8), (9) and Cauchy–Schwarz,

$$\frac{A}{2} \le A_{\mathrm{hi}} \le 3\sqrt{E} \left(\int_0^t f'^4 w(f)^2 \, \mathrm{d}s \right)^{1/2} \le 3\sqrt{E}\sqrt{A} L.$$

Division is legitimate because $A > 0$ and $L \ge \log 4 > 0$. This proves (7). □

Suppose a control of duration $t > 0$ reaches a negative fourth-axis point. Then $f = x_3$ has all six zero endpoint jets in Lemma 2. Equations (2) and (6) give

$$x_4(t) = E - \int_0^t x_5^2 f'^4 \, \mathrm{d}s. \tag{10}$$

If $A = 0$, continuity gives $f' = 0$, and this displacement is zero. If $x_4(t) < 0$, therefore, $A > 0$ and $E < t^2 A$ by (5). The logarithmic estimate implies

$$\log \frac{t^9}{4A} > \frac{1}{6t}, \qquad A < \frac{t^9}{4} e^{-1/(6t)}, \qquad 0 < -x_4(t) < \frac{t^{11}}{4} e^{-1/(6t)}.$$

The last expression is strictly increasing in $t > 0$, so the same bound holds for every negative fourth-axis point reached in any duration at most t. This proves (4) for the entire measurable control cube, without a restriction on the number of switches.

3 Concentrating loops and a reachable neighborhood

The equation $ff'' + f'^2/3 = 0$ has the profile $f(s) = s^{3/4}$ on $s > 0$. Its critical energy vanishes, while f'^4 has a logarithmically divergent integral at zero. The following endpoint corrections turn that profile into admissible closed trajectories.

Lemma 3. *There are C^2 functions f_η on $[0,1]$, with Lipschitz second derivative and zero endpoint jets through order two, such that*

$$A_\eta := \int_0^1 f_\eta'^4 \,\mathrm{d}s = A_0 + \frac{81}{256}\log\frac{c}{\eta}, \qquad E_\eta := \int_0^1 \left(f_\eta f_\eta'' + \frac{1}{3}f_\eta'^2\right)^2 \mathrm{d}s = E_0, \tag{11}$$

for $0 < \eta < c = 1/4$, where A_0, E_0 are finite positive constants. In particular, $E_\eta/A_\eta \to 0$ as $\eta \downarrow 0$.

Proof. Set

$$h(r) = \frac{221}{32}r^3 - \frac{153}{16}r^4 + \frac{117}{32}r^5,$$
$$k(r) = 1 + \frac{3}{4}r - \frac{3}{32}r^2 - \frac{455}{32}r^3 + \frac{663}{32}r^4 - \frac{261}{32}r^5.$$

Their endpoint jets are

$$(h,h',h'')(0) = (0,0,0), \qquad (h,h',h'')(1) = (1,3/4,-3/16),$$
$$(k,k',k'')(0) = (1,3/4,-3/16), \qquad (k,k',k'')(1) = (0,0,0).$$

For $0 < \eta < c$, define

$$f_\eta(s) = \begin{cases} \eta^{3/4}h(s/\eta), & 0 \le s \le \eta, \\ s^{3/4}, & \eta \le s \le c, \\ c^{3/4}k((s-c)/c), & c \le s \le 2c, \\ 0, & 2c \le s \le 1. \end{cases} \tag{12}$$

The jets match at all three junctions. Each third derivative is bounded on its piece for fixed $\eta > 0$, and the second derivative is continuous at the junctions. Consequently f_η'' is Lipschitz and $M_\eta := \|f_\eta'''\|_{L^\infty} < \infty$.

On $[\eta, c]$, the critical expression is zero and $f_\eta'^4 = (81/256)s^{-1}$. In the two polynomial pieces the changes of variables $r = s/\eta$ and $r = (s-c)/c$ cancel the scale factors exactly. Thus (11) holds with

$$A_0 = \int_0^1 h'^4 \,\mathrm{d}r + \int_0^1 k'^4 \,\mathrm{d}r,$$
$$E_0 = \int_0^1 (hh'' + h'^2/3)^2 \,\mathrm{d}r + \int_0^1 (kk'' + k'^2/3)^2 \,\mathrm{d}r.$$

These constants are finite. Also $A_0 > 0$ since h' is not identically zero. If $a_0 = 221/32$, then $hh'' + h'^2/3 = 9a_0^2 r^4 + O(r^5)$, so $E_0 > 0$. Finally, the logarithm in (11) diverges as $\eta \downarrow 0$. □

Proof of Theorem 1. The polynomial vector fields, global solutions and negative-axis estimate were established in Section 2. We construct a full neighborhood, fixing an arbitrary $t > 0$. Let $\sigma = t/4$ and $a = t/16$. By Lemma 3, choose $\eta_- \in (0, c)$ with $E_0 < a^2 A_{\eta_-}$, and set

$$\beta_-^{\max} = \frac{\sigma^3}{1 + M_{\eta_-}}.$$

First apply $u = 0, v = 1$ for a duration a. The first three coordinates remain zero and $x_5 = a$; there is no fourth displacement. Keep $v = 0$ during the next interval of length σ, and in its local time s use

$$u(s) = \beta\sigma^{-3} f'''_{\eta_-}(s/\sigma), \qquad 0 \le \beta \le \beta_-^{\max}.$$

This measurable control has absolute value at most one. The triple coordinates are

$$(x_1, x_2, x_3)(s) = \big(\beta\sigma^{-2} f''_{\eta_-}(s/\sigma), \beta\sigma^{-1} f'_{\eta_-}(s/\sigma), \beta f_{\eta_-}(s/\sigma)\big),$$

so they return to zero. The fourth displacement is exactly

$$\beta^4\sigma^{-3}(E_0 - a^2 A_{\eta_-}). \tag{13}$$

Apply $u = 0, v = -1$ for another duration a to return x_5 to zero. This also gives no fourth displacement. The total duration is $3t/8$, and varying β fills an interval $[-d_-, 0]$ with $d_- > 0$.

For a positive loop take $\eta_+ = c/2$ and keep $v = 0$. The same triple trajectory, now with $0 \le \beta \le \beta_+^{\max} := \sigma^3/(1 + M_{\eta_+})$, gives fourth displacement $\beta^4\sigma^{-3}E_0$ in time σ. It fills $[0, d_+]$ for some $d_+ > 0$. Both intervals are filled exactly, since every nonnegative real number has a nonnegative fourth root. The other four coordinates end at zero. These loops add the same fourth displacement from any initial x_4.

It remains to steer the first three and fifth coordinates while compensating for their fourth displacement. Put $\ell = t/4$. For the triple integrator from zero, its endpoint under a control u is

$$\left(\int_0^\ell u\,\mathrm{d}s, \int_0^\ell (\ell - s)u\,\mathrm{d}s, \int_0^\ell \frac{(\ell - s)^2}{2} u\,\mathrm{d}s\right). \tag{14}$$

Let $p_0(s) = 1$, $p_1(s) = \ell - s$, $p_2(s) = (\ell - s)^2/2$ and $G_{ij} = \int_0^\ell p_i p_j\,\mathrm{d}s$ for $0 \le i, j \le 2$. The Gram matrix G is positive definite because these polynomials are linearly independent. For $z = (z_1, z_2, z_3, z_5)$, choose

$$u_z(s) = \sum_{i=0}^{2} \big(G^{-1}(z_1, z_2, z_3)^{\mathsf{T}}\big)_i p_i(s), \qquad v_z(s) = \frac{z_5}{\ell}.$$

They give the endpoint z in the four specified coordinates. For fixed ℓ, their sup norms depend continuously on z and vanish at zero. Thus the controls are admissible for all sufficiently small z. Their fourth displacement $F(z)$ is a polynomial in z, with terms of degrees four and six, because the first three coordinates and x_5 depend linearly on z. In particular, F is continuous and $F(0) = 0$.

Set $d = \min(d_-, d_+) > 0$. Choose $\delta > 0$ so small that for $|y| < \delta$, the controls associated with $z = (y_1, y_2, y_3, y_5)$ are admissible and

$$|y_4 - F(z)| < d.$$

Such a choice follows from continuity at zero, for example by imposing $|y_4| < d/2$ and $|F(z)| < d/2$. Execute the scalar loop with displacement $y_4 - F(z)$ first, and then apply u_z, v_z for a duration ℓ. The final point is exactly y: the second stage starts with the other four coordinates zero, and its fourth displacement is independent of the initial x_4. The total duration is at most $3t/8 + t/4 = 5t/8 < t$. This proves (3).

Finally, write $b(t) = t^{11}e^{-1/(6t)}/4$. For every integer $N \ge 1$, $b(t)/t^N \to 0$ as $t \downarrow 0$. Given $C, T > 0$, choose $0 < t < T$ with $b(t) < Ct^N/2$. The point $-(C/2)t^N e_4$ lies in $B(0, Ct^N)$ but is excluded from $\mathcal{R}_{\le t}(0)$ by (4). This proves the failure of every polynomial inner ball and completes the counterexample. □

AI-use statement. OpenAI Codex was used for literature retrieval, mathematical development, exact arithmetic checks, and manuscript drafting. Independent mathematical review is requested.

References

[1] S. Jafarpour, *On small-time local controllability*, SIAM Journal on Control and Optimization **58** (2020), 425–446. doi:10.1137/16M1068797. Preprint PDF cited for Conjecture 1.2 (p. 3): arXiv:1604.02432v7 (2019).

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[3] K. Beauchard and F. Marbach, *Some quartic control results for scalar-input systems*, arXiv:2608.28582v1 (2026).

[4] J. K. Hunter, *Notes on Partial Differential Equations*, Appendix 3.A, Absolutely continuous functions, Theorems 3.60 and 3.62, p. 79.